NADYAH RIZKY AMELIA

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A fresh graduate in Telecommunication Engineering from Telkom University with a GPA of 3.73/4.00 and cum laude honors. During an internship at Bank Indonesia, assisted in reviewing IT infrastructure and network architecture documentation submitted by banks participating in BI-FAST to support compliance monitoring, alongside processing, verifying, and analyzing BI-RTGS transaction summaries for internal reporting. At PT Infomedia Nusantara, supported Telkomsel call center service evaluation by reviewing inbound call and agent performance data. Applied technical knowledge through a final-year project (Capstone Design) titled “Design of a Machine Learning-Based Network Intrusion Prevention System for Zero-Day Attack Detection with Monitoring Features.” The work involved developing a machine learning-based system using Snort 3 and evaluating attack detection, traffic blocking, and network performance, with InfluxDB and Grafana integrated for monitoring. Related research was presented as first author at ICITACEE 2026. Beyond technical work, involvement in laboratory teaching and student organizations strengthened communication, teamwork, and problem-solving skills. Technical skills include network configuration, monitoring, and basic troubleshooting, with hands-on experience in Linux and virtual network environments, complemented by MikroTik Certified Network Associate (MTCNA) certification. Interested in applying this background to network engineering, network operations, and network security to support reliable and secure network services.

PENGALAMAN KERJA

Bank Indonesia - Kota Adm. Jakarta Pusat, DKI Jakarta, Indonesia **Juni 2025 - Agustus 2025**

Internship

Supported the KIM Division within the Departemen Penyelenggaraan Sistem Pembayaran (DPSP), Bank Indonesia. Responsibilities included: (1) Reviewing IT infrastructure and network architecture documentation submitted by BI-FAST participant banks against PADG No. 14 of 2025 requirements and documenting findings for compliance monitoring. (2) Processing, verifying, and analyzing BI-RTGS transaction summaries to support internal monitoring and reporting. (3) Preparing and presenting materials on cybersecurity issues in the banking sector, including DDoS, man-in-the-middle attacks, and SQL injection, for an internal staff discussion.

telkom university - Kab. Bandung, Jawa Barat, Indonesia **Juni 2023 - Juni 2025**

Practicum Assistant

(1) Guided more than 100 students from 10 study programs through Basic Physics practicum sessions over four semesters, explaining concepts and demonstrating experimental procedures. (2) Entered and analyzed data for more than 50 practicum tools and managed student grade records to support practicum documentation and assessment. (3) Conducted research on practicum tool operation and maintained more than 50 tools to support laboratory sessions.

PT. Infomedia Nusantara - Kota Bandung, Jawa Barat, Indonesia **Januari 2025 - Februari 2025**

Internship

(1) Reviewed Telkomsel inbound call data to support call center operational monitoring and identify potential service issues. (2) Reviewed call center agent performance data and documented findings for internal evaluation.

PENDIDIKAN

S1 - Universitas Telkom, Kab. Bandung, Jawa Barat, Indonesia **2022 - 2026**

TEKNIK TELEKOMUNIKASI - 3.73

KETERAMPILAN

Problem Solving, Network Configuration, Teamwork, Technical Documentation, Network monitoring, Network Troubleshooting, analytical thinking, Computer networking, Data Analysis, Network Security.

SERTIFIKASI

MikroTik Certified Network Associate (MTCNA) **Oktober 2026 - Oktober 2029**

MikroTik (Mikrotikls SIA)

PELATIHAN

Teknologi Informasi Dan Komunikasi - MikroTik Certified Network Associate (MTCNA) Training **Oktober 2026 - Oktober 2026**

Id Networkers

Teknologi Informasi Dan Komunikasi - Networking Basics **September 2026 - September 2026**

Cisco Networking Academy

Teknologi Informasi Dan Komunikasi - Introduction to Cybersecurity **Februari 2026 - Februari 2026**

Cisco Networking Academy

PENCAPAIAN

Analisa dan Simulasi Perancangan Jaringan FTTH dengan Menggunakan Teknologi XGPON di Perumahan Cluster Anapuri – Podomoro Park Bandung **2026**

Co-authored a research article published in TEKTRIKA, Vol. 11 No. 1 (2026), examining the design and simulation of an FTTH network using XG-PON technology in Cluster Anapuri, Podomoro Park Bandung. The study used OptiSystem to assess transmission quality and network feasibility through optical attenuation, received power, Bit Error Rate (BER), and Q-factor analysis. Simulation results showed total attenuation of 21.56 dB, a Q-factor of 7.0141, and a minimum BER of approximately 1.01 × 10?¹². These findings supported the feasibility of the proposed network under the simulated conditions, with potential applications in residential internet connectivity, smart home systems, IoT devices, and multimedia services.

Experimental Evaluation of an Inline Snort 3 NIPS with AE-MLP-Based Detection **2026**

Authored and presented this research as first author at the 13th International Conference on Information Technology, Computer, and Electrical Engineering (ICITACEE 2026). The paper evaluates an inline Snort 3 network intrusion prevention system integrating AE-MLP detection, temporary IP blocking, and monitoring through InfluxDB and Grafana. Five attack scenarios were tested across three independent runs, with MLP and LightGBM included as comparison models. Network performance and resource utilization were also evaluated under controlled vCPU and memory variations, with 0% packet loss recorded during UDP forwarding tests at 10 and 25 Mbit/s.

2nd Place – Business Case Project Contest, ACAVERO 2024 **2024**

Awarded second place at ACAVERO 2024 as part of the Silent Grex team for proposing “ABC Control,” a centralized ERP solution for a global electronic components supplier. The business case addressed fragmented legacy systems, inventory imbalances, and difficulties tracking product quality. Supported by SWOT analysis, the proposal integrated warehouse management, purchasing and supply chain management, and supplier relationship management. Recommended features included barcode-based inventory tracking, centralized production and quality monitoring, and a phased implementation plan with employee training to improve data visibility and operational coordination.